



2022 ANNUAL EXAMINATION PHYSICS Form V

STRUCTURE OF PAPER

Section I: Multiple Choice (14 marks)

Answer all questions on the Multiple-Choice Answer Sheet.

Section II: Extended Response (76 marks)

Answer all parts of the questions in the spaces provided in the Examination Paper.

Board approved calculators may be used

EXAMINATION

Date: Friday 2nd September 8.40am
Duration: 2 hours
Marks: 90

CHECKLIST

In the stapled bundle each boy should have the following:

- Examination sections
- 1 sheet extra writing paper
- Data and Formula Sheets
- Multiple-Choice Answer Sheet

EXAM INSTRUCTIONS

Remove the centre staple and hand in all parts of the examination in a neat bundle.

WRITE YOUR CANDIDATE NUMBER IN THE SPACE PROVIDED AT THE TOP OF EACH SEPARATE SECTION.

There is a Data/Formula sheet included at the end of the paper.

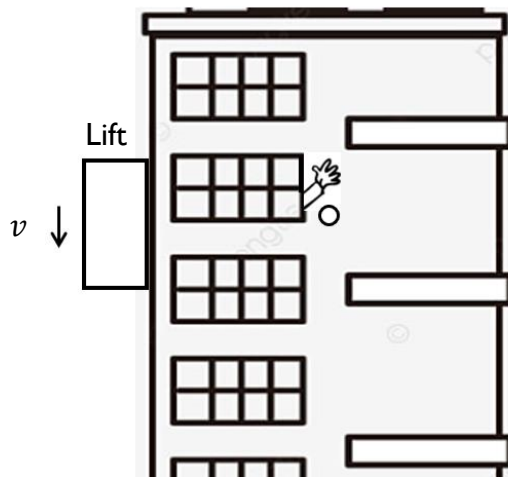
Responses requiring more writing space should be clearly be marked **CONTINUED**. Additional extra writing space is provided at the end of the exam. When the response is continued on extra writing paper it should clearly indicate the question number.

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SECTION I: MULTIPLE CHOICE

Attempt ALL Questions
Use the Multiple-Choice Answer Sheet.

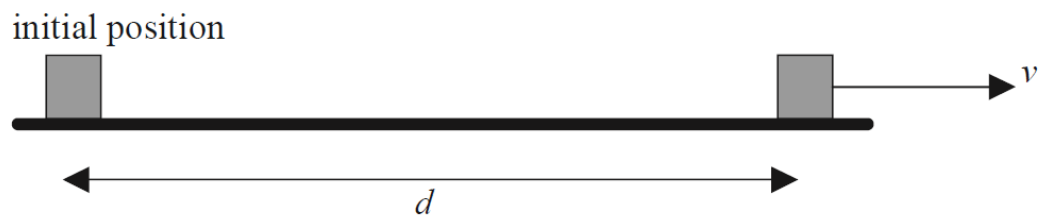
- 1 A ball is thrown vertically upwards. As it passes its highest point, which of the following quantities will change sign?
- (A) Displacement
 - (B) Velocity
 - (C) Acceleration
 - (D) Force
- 2 A lift is moving down the side of a building at constant speed v . As it passes a window, a ball is dropped from rest.



Just after the ball is released, the velocity of the lift relative to the ball will be

- (A) Downwards and decreasing in magnitude.
- (B) Downwards and increasing in magnitude.
- (C) Upwards and decreasing in magnitude.
- (D) Upwards and increasing in magnitude.

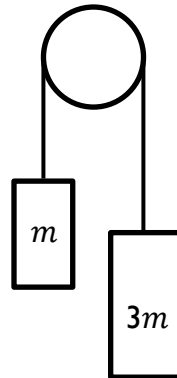
- 3 A train accelerates uniformly from rest at a rate of 0.28 m s^{-2} over a distance of 400 m. What time did it take to cover that distance?
- (A) 38 s
(B) 53 s
(C) 56 s
(D) 112 s
- 4 A box is accelerated from rest at a constant rate. When it is a distance d from its initial position it is moving with a velocity v .



When it was a distance $\frac{d}{2}$ from its initial position its velocity was

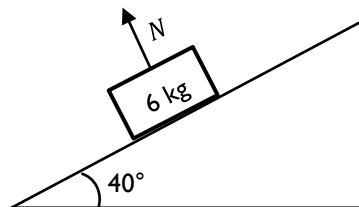
- (A) $\frac{v}{4}$
(B) $\frac{v}{2\sqrt{2}}$
(C) $\frac{v}{2}$
(D) $\frac{v}{\sqrt{2}}$

- 5 The diagram shows two masses joined by a lightweight string over a pulley.



What is the magnitude of the acceleration of the system?

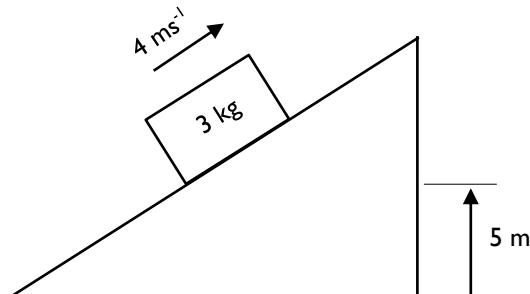
- (A) 0.5 m s^{-2}
 - (B) 4.9 m s^{-2}
 - (C) 6.5 m s^{-2}
 - (D) 9.8 m s^{-2}
- 6 The diagram shows a stationary 6 kg block on a ramp.



What is the magnitude of the normal force on the block?

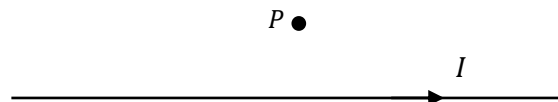
- (A) 4.6 N
- (B) 37.8 N
- (C) 45.0 N
- (D) 58.8 N

- 7 A block is on a frictionless slope, initially 5 m vertically above the ground. It is given an initial velocity of 4 m s^{-1} and slides up the slope before eventually stopping and returning down the slope until it reaches the bottom.



What is the velocity of the block when it reaches the bottom?

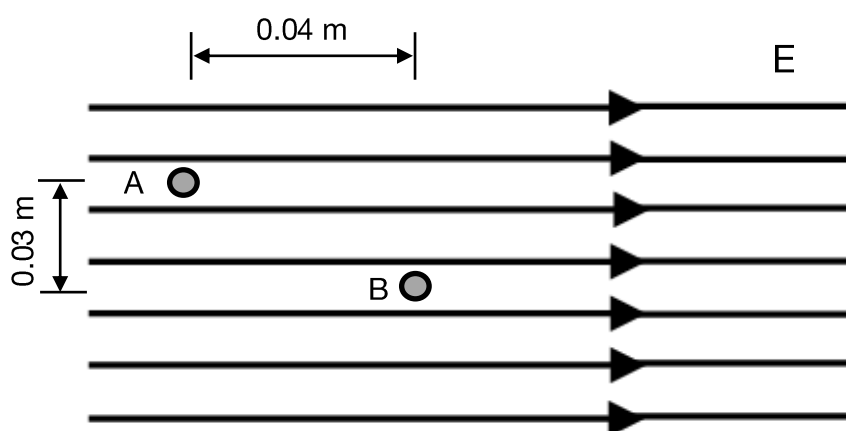
- (A) 10.7 m s^{-1}
 - (B) 12.1 m s^{-1}
 - (C) 24 m s^{-1}
 - (D) 114 m s^{-1}
- 8 A wire carries a current as shown.



If a compass was placed at point P , the compass needle would point

- (A) to the left.
- (B) to the right.
- (C) into the page.
- (D) out of the page.

- 9 A charge of $7.0 \times 10^{-6} \text{ C}$ is moved from position A to position B within a uniform electric field of strength 2000 N C^{-1} .



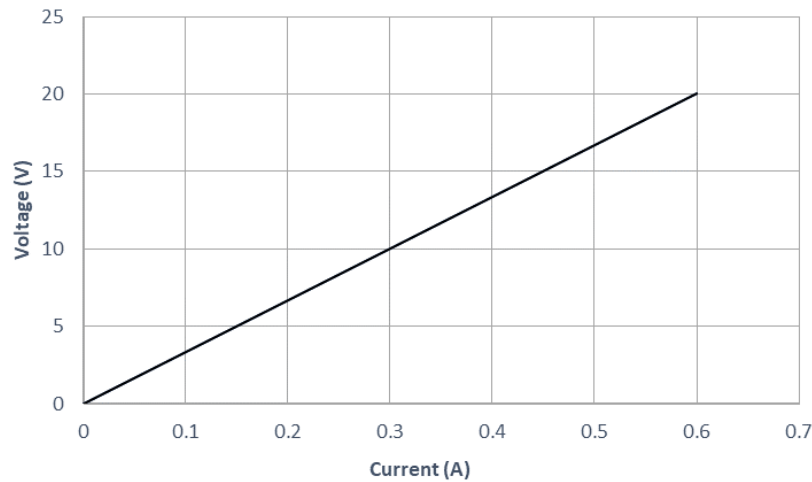
What will be the change in energy of the charge?

- (A) $2.8 \times 10^{-4} \text{ J}$
 - (B) $4.2 \times 10^{-4} \text{ J}$
 - (C) $5.6 \times 10^{-4} \text{ J}$
 - (D) $7.0 \times 10^{-4} \text{ J}$
- 10 A radio wave of frequency 300 MHz would have a wavelength of
- (A) 0.1 m
 - (B) 1.0 m
 - (C) 10 m
 - (D) $1.0 \times 10^6 \text{ m}$
- 11 A light ray that has a wavelength of 600 nm in air travels from the air into glass with refractive index of 1.5 .

Determine the frequency of the light ray in the glass.

- (A) $3.3 \times 10^{11} \text{ Hz}$
- (B) $5.0 \times 10^{11} \text{ Hz}$
- (C) $3.3 \times 10^{14} \text{ Hz}$
- (D) $5.0 \times 10^{14} \text{ Hz}$

- 12 The graph below shows the voltage-current characteristics of a resistor.



Calculate the resistance of the resistor.

- (A) $0.03 \, \Omega$
(B) $12 \, \Omega$
(C) $20 \, \Omega$
(D) $33 \, \Omega$
- 13 When X identical resistors are wired in series, their total resistance is 64 times bigger than when they are wired in parallel. How many resistors are there?
- (A) 4
(B) 8
(C) 16
(D) 64
- 14 In a particle accelerator, bunches of high-speed protons travel around a ring, forming a current of 0.58 amps. If there are 3.12×10^7 bunches passing per second, how many protons are in each bunch?
- (A) 1.2×10^{11}
(B) 1.8×10^{11}
(C) 3.6×10^{18}
(D) 3.4×10^{26}

SECTION II: Part A (19 Marks)

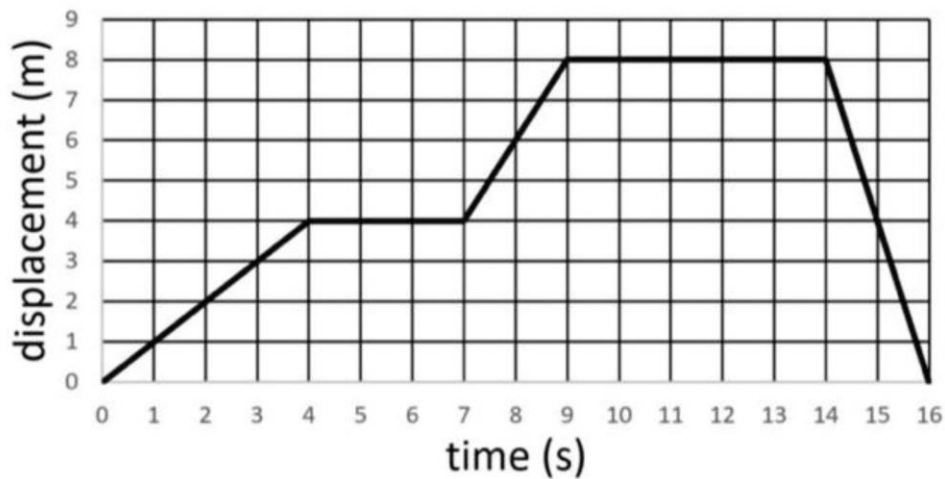
Answer the questions in the spaces provided.
Show all relevant working in questions involving calculations.

CANDIDATE NUMBER							

Question 15 (3 marks)

Marks

The graph below shows the displacement of a remote-controlled car as it moves along a straight line.



- (a) What is the total distance traveled by the car?

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- (b) What is the maximum speed of the car?

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- (c) What is the average velocity of the car between 0 and 14 s?

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Question 16 (4 marks)**Marks**

A car drives from one house to another along the path shown below.



- (a) If the trip takes 20 minutes, calculate the average speed of the car.

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- (b) Calculate the displacement of the car from its starting point (magnitude and direction).

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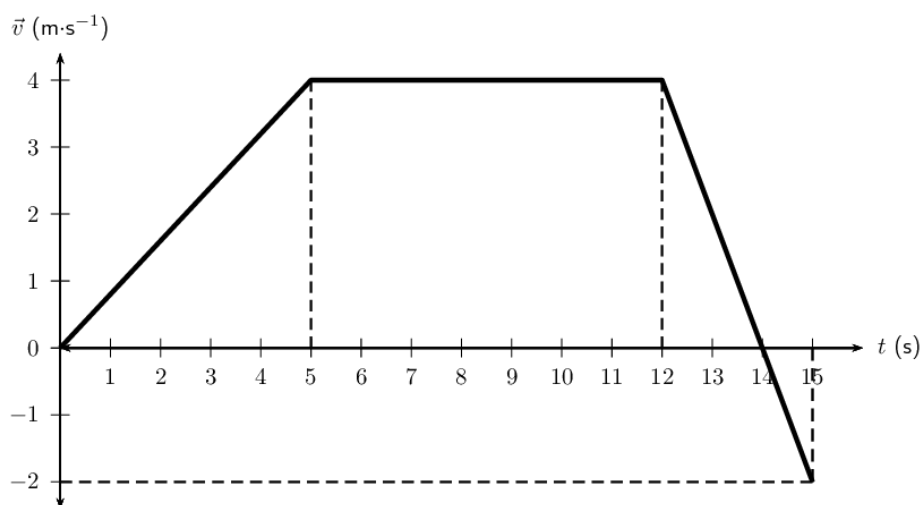
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Question 17 (4 marks)**Marks**

A cyclist is moving along a straight road, initially heading west. The graph of his velocity as a function of time is shown.



- (a) Identify a time at which the bike was stationary.

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- (b) Calculate the magnitude of the maximum acceleration.

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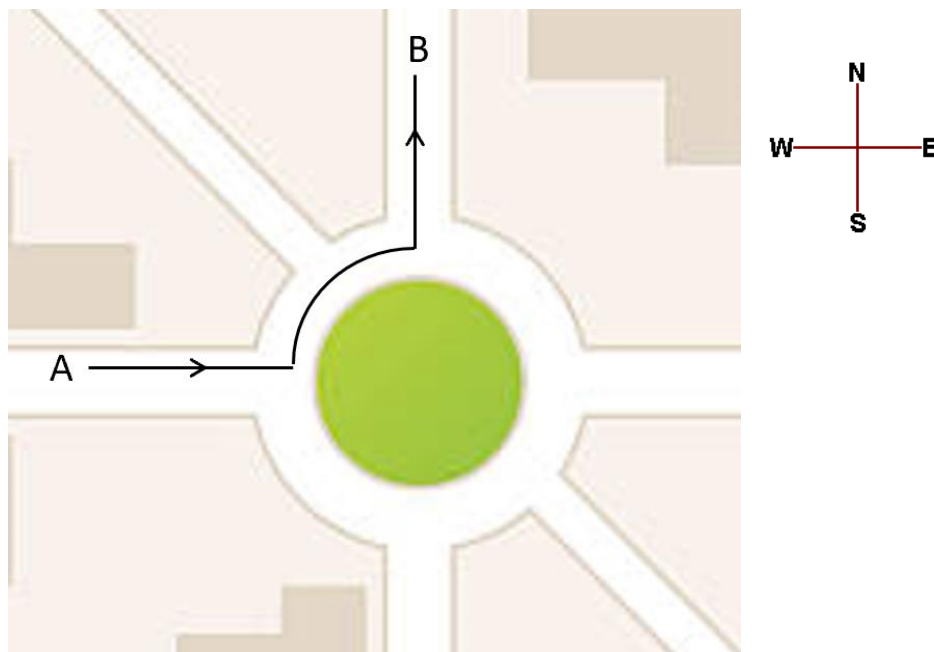
- (c) Calculate the displacement after 15 s.

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Question 18 (3 marks)**Marks**

A car uses a roundabout to make a left turn, following the path shown below. The circular section of the car's path has a radius of 16m and the car maintains a constant speed from A to B.



- (a) If the car takes 4.6 s to complete the circular section of the road, calculate its speed.

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- (b) Calculate the change in velocity from A to B (magnitude and direction).

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Question 19 (5 marks)**Marks**

A plane accelerates from rest down a runway that is 2000 m long. It can accelerate at 3 m s^{-2} and decelerate at 2 m s^{-2} .

There is a point of no return when the plane cannot stop before the end of the runway if it fails to take off.

- (a) On the axes below, sketch a velocity-time graph for the situation where the plane fails to take off but uses the entire length of the runway (no numerical values required).

**2**

- (b) Calculate the maximum speed reached by the plane in that situation.

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Section II extra writing space

If you use this space, clearly indicate which question you are answering.

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SECTION II: Part B (15 Marks)

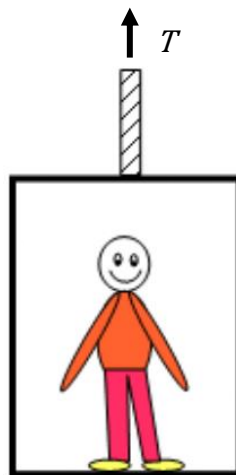
Answer the questions in the spaces provided.
Show all relevant working in questions involving calculations.

CANDIDATE NUMBER							

Question 20 (3 marks)

Marks

A 70 kg man is standing in a lift of mass 800 kg, as shown. The lift is accelerating upwards at 1.5 m s^{-2} .



- (a) Determine the magnitude of the normal force of the lift on the man.

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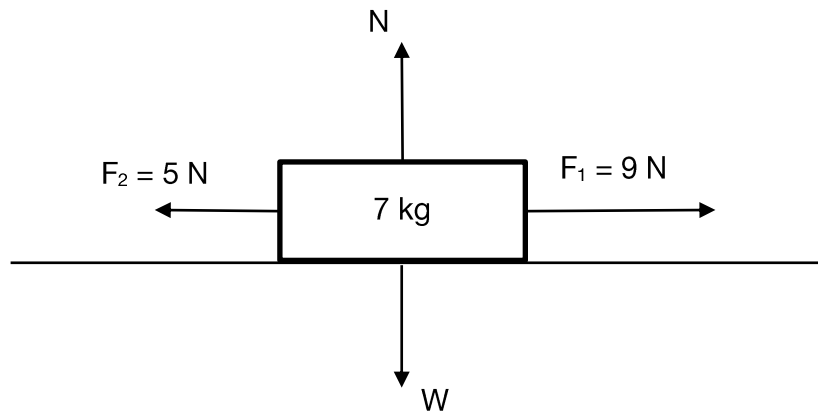
- (b) Determine the magnitude of the tension T in the lift rope.

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2

Question 21 (5 marks)**Marks**

A 7 kg object is pulled by an applied force $F_1 = 9\text{ N}$ and moves to the right. There is a friction force $F_2 = 5\text{ N}$ acting against its direction of motion.



- (a) Determine the magnitude of the normal force on the block.

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- (b) Determine the acceleration of the block.

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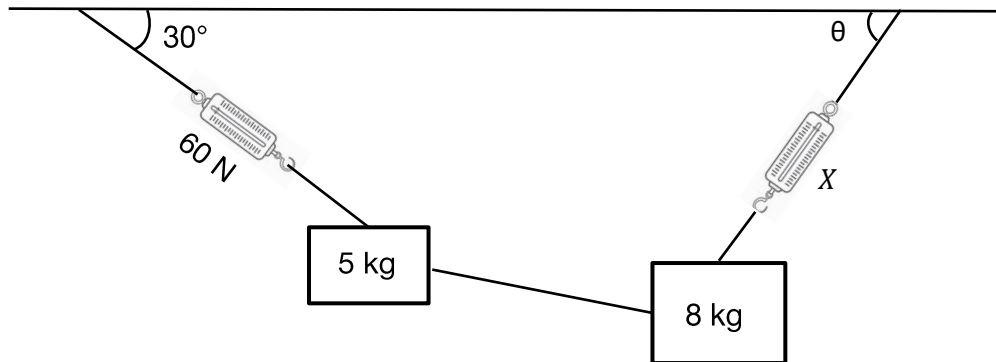
- (c) Determine the coefficient of kinetic friction between the block and the surface.

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Question 22 (4 marks)**Marks**

Two masses are suspended at rest from a ceiling by two spring balances, one of which has unknown tension X . The masses are attached together as pictured below.



- (a) Determine the horizontal component of the 60 N force.

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- (b) Determine the angle θ .

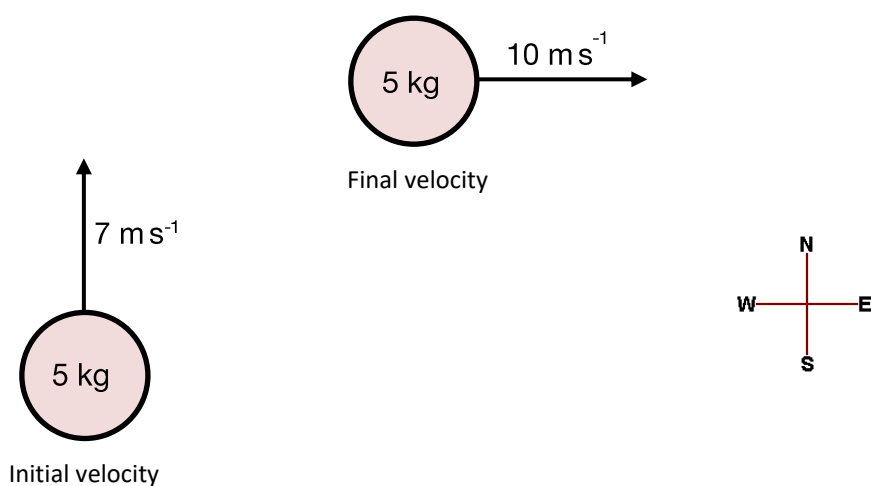
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Question 23 (3 marks)**Marks**

A 5 kg object is travelling initially north at 7 m s^{-1} when a constant force is applied to it for 4 seconds.

After 4 seconds the object is travelling at 10 m s^{-1} east.



Calculate the magnitude and direction of the force that is required to change the object's motion as described.

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SECTION II: Part C (15 Marks)

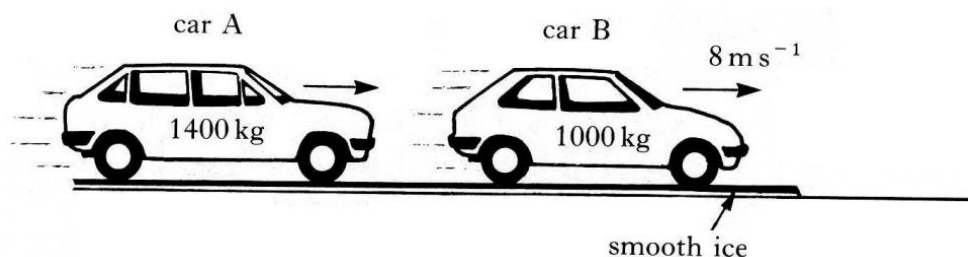
Answer the questions in the spaces provided.
Show all relevant working in questions involving calculations.

CANDIDATE NUMBER							

Question 24 (5 marks)

Marks

Two cars travelling in the same direction but at different speeds, skid on a patch of smooth ice, as shown below.



Car A, of mass 1400 kg, skids straight into the back of Car B, of mass 1000 kg. Immediately before the impact, Car B is moving with a speed of 8 m s^{-1} . The two cars become entangled after the impact and continue to move in the same straight line.

Immediately after the impact, the speed of the entangled cars is 15 m s^{-1} .

- (a) Determine the momentum of Car B just before the collision.

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- (b) Determine the speed of Car A just before the collision.

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- (c) After the collision, the cars leave the patch of ice and continue skidding along the rough road. They come to rest in a distance of 20 m after leaving the ice. Calculate the magnitude of the average frictional force acting on the cars as they come to rest.

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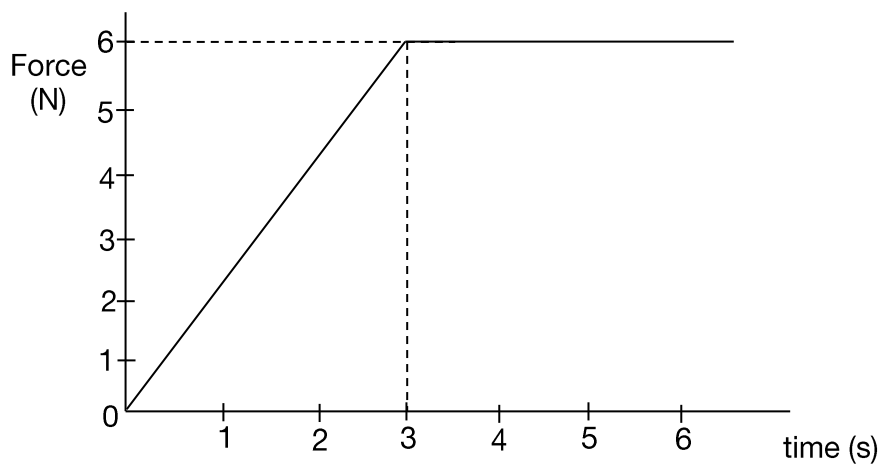
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Question 25 (4 marks)**Marks**

The graph below shows how the force on a 2.4 kg object varies with time. The object's velocity was initially 10 m s^{-1} east and the force was applied in a westerly direction.



- (a) Calculate the object's change in momentum during the first 3 seconds (magnitude and direction).

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- (b) Determine the time at which the object is at rest.

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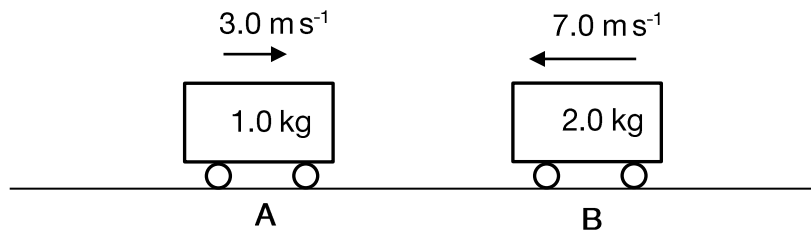
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Question 26 (3 marks)**Marks**

Trolley A of mass 1.0 kg is moving with a velocity of 3.0 m s^{-1} towards the right. It collides with trolley B of mass 2.0 kg moving left with a velocity of 7.0 m s^{-1} .



If the collision is perfectly elastic, what is the final velocity of the trolley A?

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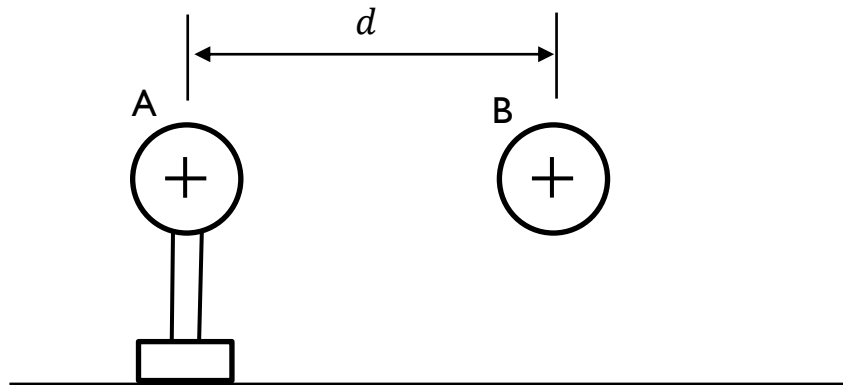
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Question 27 (3 marks)**Marks**

A positively charged particle A is fixed on a stand. Another positive charge B is held a distance d away it, as shown. Assume no external forces act on the particles.



At $t = 0$, particle B is released. Describe what changes will subsequently occur to the motion of particle B and the force between the particles, and justify your answer.

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SECTION II: Part D (15 Marks)

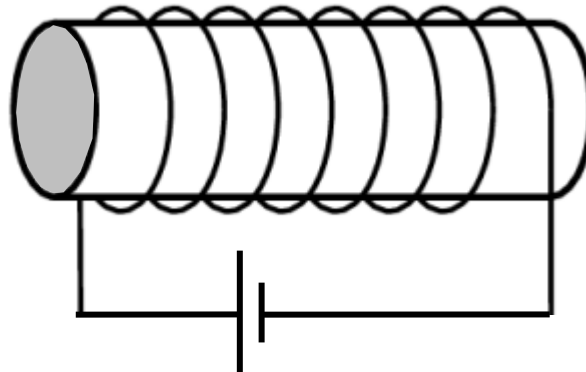
Answer the questions in the spaces provided.
Show all relevant working in questions involving calculations.

CANDIDATE NUMBER							

Question 28 (3 marks)

Marks

A coil of wire is wrapped around a hollow cardboard tube, as shown. A power supply is connected to the coil and the current creates an electromagnet.



The coil carries a current of 6 A and has a length of 15 cm, with 40 turns of wire.

- (a) Determine the strength of the magnetic field inside the coil.

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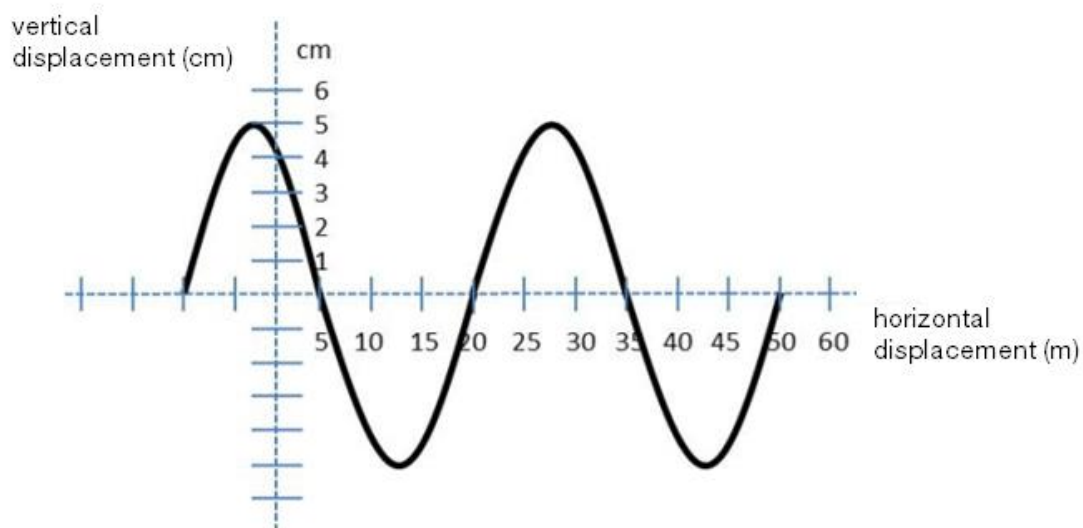
2

- (b) Indicate on the diagram the position of the North pole of the electromagnet.

1

Question 29 (4 marks)**Marks**

The graph below shows vertical displacement versus horizontal displacement for a wave travelling to the right at a speed of 10 m s^{-1} .



From the information in the graph, determine:

- (a) the wavelength of the wave.

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- (b) the amplitude of the wave.

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- (c) the period of the wave.

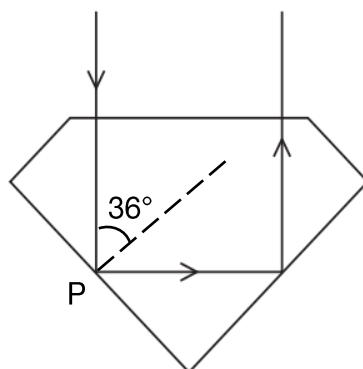
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Question 30 (4 marks)

Marks

A ray of light enters a gemstone as shown below. It reflects twice within the stone, then exits.



- (a) Show that the refractive index of the stone must be at least 1.7. Justify your answer.

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- (b) Assuming that the refractive index of the gemstone is 1.8, determine the speed of the light within the stone.

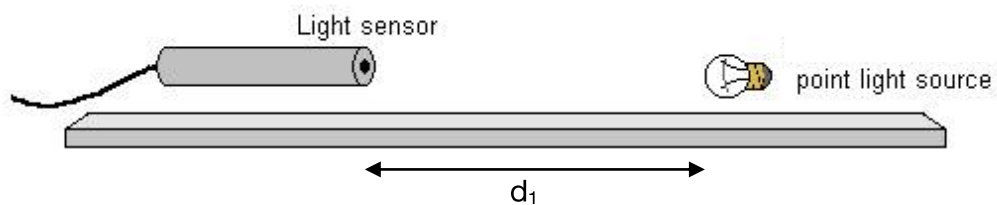
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Question 31 (2 marks)**Marks**

A light sensor which measures the intensity of light is placed a distance $d_1 = 15 \text{ cm}$ from a point light source. If the intensity at that distance is 2.0 W m^{-2} , determine the distance d_2 at which the intensity will be 1.5 W m^{-2} .



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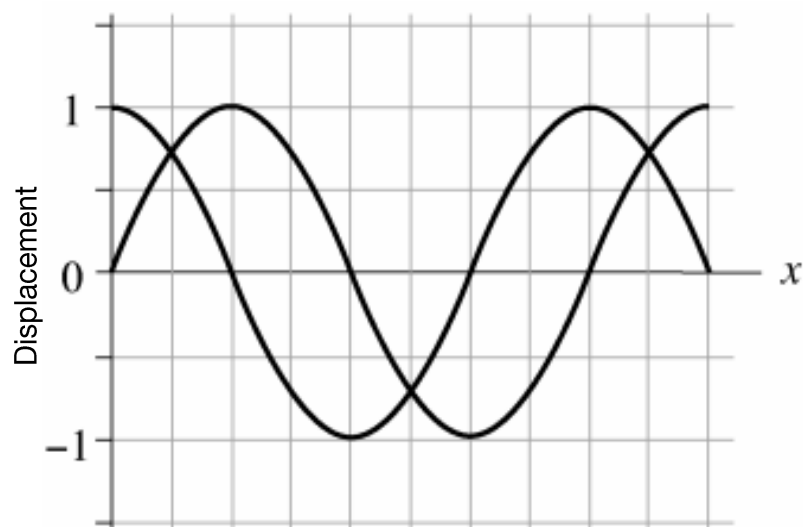
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2**Question 32 (2 marks)**

Two waves are depicted on the graph below. Draw on the graph the superposition of the two waves.

**2**

SECTION II: Part E (12 Marks)

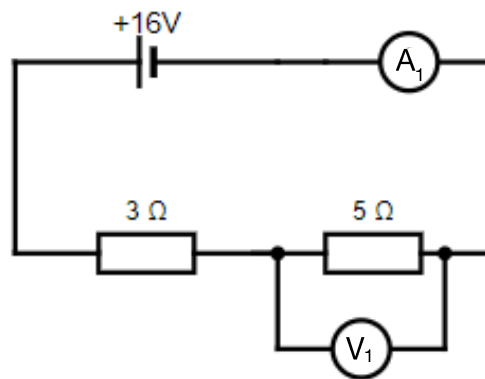
Answer the questions in the spaces provided.
Show all relevant working in questions involving calculations.

CANDIDATE NUMBER							

Question 33 (2 marks)

Marks

Calculate the readings on the ammeter and voltmeter in the circuit below.



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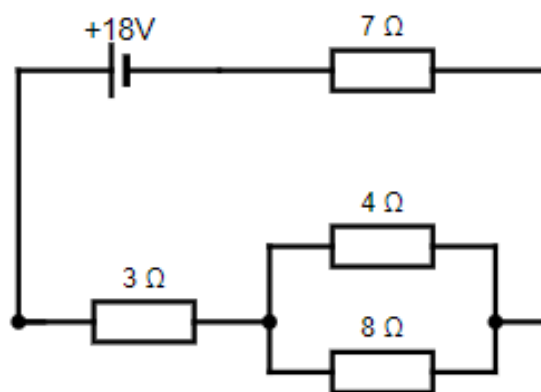
$A_1 = \dots\dots\dots$

$V_1 = \dots\dots\dots$

2

Question 34 (5 marks)**Marks**

Consider the circuit shown below.



Calculate:

- (a) The total resistance of the circuit.

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- (b) The current through the 3 Ω resistor.

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- (c) The voltage across the 8 Ω resistor.

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- (d) The current through the 4 Ω resistor.

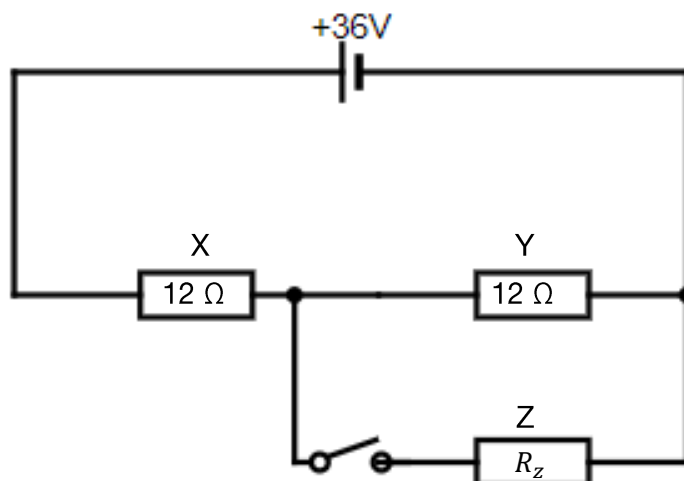
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Question 35 (5 marks)**Marks**

Consider the circuit shown below.



- (a) Calculate the amount of power used by Resistor X when the switch is open.

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- (b) When the switch is closed, the total power consumption of the whole circuit increases by a factor of $\frac{5}{3}$. Calculate the resistance R_Z of Resistor Z.

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3**END OF EXAMINATION**

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Section II extra writing space

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Section II extra writing space

If you use this space, clearly indicate which question you are answering.

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Physics

DATA SHEET

Charge on electron, q_e	$-1.602 \times 10^{-19} \text{ C}$
Mass of electron, m_e	$9.109 \times 10^{-31} \text{ kg}$
Mass of neutron, m_n	$1.675 \times 10^{-27} \text{ kg}$
Mass of proton, m_p	$1.673 \times 10^{-27} \text{ kg}$
Speed of sound in air	340 m s^{-1}
Earth's gravitational acceleration, g	9.8 m s^{-2}
Speed of light, c	$3.00 \times 10^8 \text{ m s}^{-1}$
Electric permittivity constant, ϵ_0	$8.854 \times 10^{-12} \text{ A}^2 \text{ s}^4 \text{ kg}^{-1} \text{ m}^{-3}$
Magnetic permeability constant, μ_0	$4\pi \times 10^{-7} \text{ N A}^{-2}$
Universal gravitational constant, G	$6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
Mass of Earth, M_E	$6.0 \times 10^{24} \text{ kg}$
Radius of Earth, r_E	$6.371 \times 10^6 \text{ m}$
Planck constant, h	$6.626 \times 10^{-34} \text{ J s}$
Rydberg constant, R (hydrogen)	$1.097 \times 10^7 \text{ m}^{-1}$
Atomic mass unit, u	$1.661 \times 10^{-27} \text{ kg}$ $931.5 \text{ MeV}/c^2$
1 eV	$1.602 \times 10^{-19} \text{ J}$
Density of water, ρ	$1.00 \times 10^3 \text{ kg m}^{-3}$
Specific heat capacity of water	$4.18 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$
Wein's displacement constant, b	$2.898 \times 10^{-3} \text{ m K}$

FORMULAE SHEET

Motion, forces and gravity

$$s = ut + \frac{1}{2}at^2$$

$$v^2 = u^2 + 2as$$

$$\Delta U = mg\Delta h$$

$$P = \frac{\Delta E}{\Delta t}$$

$$\sum \frac{1}{2}mv_{\text{before}}^2 = \sum \frac{1}{2}mv_{\text{after}}^2$$

$$\Delta \vec{p} = \vec{F}_{\text{net}} \Delta t$$

$$\omega = \frac{\Delta \theta}{t}$$

$$\tau = r_{\perp} F = rF \sin \theta$$

$$v = \frac{2\pi r}{T}$$

$$U = -\frac{GMm}{r}$$

$$v = u + at$$

$$\vec{F}_{\text{net}} = m\vec{a}$$

$$W = F_{\parallel} s = Fs \cos \theta$$

$$K = \frac{1}{2}mv^2$$

$$P = F_{\parallel} v = Fv \cos \theta$$

$$\sum m\vec{v}_{\text{before}} = \sum m\vec{v}_{\text{after}}$$

$$a_c = \frac{v^2}{r}$$

$$F_c = \frac{mv^2}{r}$$

$$F = \frac{GMm}{r^2}$$

$$\frac{r^3}{T^2} = \frac{GM}{4\pi^2}$$

Waves and thermodynamics

$$v = f\lambda$$

$$f = \frac{1}{T}$$

$$d \sin \theta = m\lambda$$

$$n_x = \frac{c}{v_x}$$

$$I = I_{\text{max}} \cos^2 \theta$$

$$Q = mc\Delta T$$

$$f_{\text{beat}} = |f_2 - f_1|$$

$$f' = f \frac{(v_{\text{wave}} + v_{\text{observer}})}{(v_{\text{wave}} - v_{\text{source}})}$$

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

$$\sin \theta_c = \frac{n_2}{n_1}$$

$$I_1 r_1^2 = I_2 r_2^2$$

$$\frac{Q}{t} = \frac{kA\Delta T}{d}$$

FORMULAE SHEET (continued)

Electricity and magnetism

$$E = \frac{V}{d}$$

$$V = \frac{\Delta U}{q}$$

$$W = qV$$

$$W = qEd$$

$$B = \frac{\mu_0 I}{2\pi r}$$

$$B = \frac{\mu_0 NI}{L}$$

$$\Phi = B_{\parallel} A = BA \cos \theta$$

$$\mathcal{E} = -N \frac{\Delta \Phi}{\Delta t}$$

$$\frac{V_p}{V_s} = \frac{N_p}{N_s}$$

$$\vec{F} = q\vec{E}$$

$$F = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2}$$

$$I = \frac{q}{t}$$

$$V = IR$$

$$F = qv_{\perp} B = qvB \sin \theta$$

$$F = I_{\perp} B = I B \sin \theta$$

$$\frac{F}{l} = \frac{\mu_0}{2\pi} \frac{I_1 I_2}{r}$$

$$\tau = nIA_{\perp} B = nIAB \sin \theta$$

$$V_p I_p = V_s I_s$$

Quantum, special relativity and nuclear

$$\lambda = \frac{h}{mv}$$

$$K_{\max} = hf - \phi$$

$$\lambda_{\max} = \frac{b}{T}$$

$$E = mc^2$$

$$E = hf$$

$$\frac{1}{\lambda} = R \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$$

$$t = \frac{t_0}{\sqrt{\left(1 - \frac{v^2}{c^2}\right)}}$$

$$l = l_0 \sqrt{\left(1 - \frac{v^2}{c^2}\right)}$$

$$p_v = \frac{m_0 v}{\sqrt{\left(1 - \frac{v^2}{c^2}\right)}}$$

$$N_t = N_0 e^{-\lambda t}$$

$$\lambda = \frac{\ln 2}{t_{\frac{1}{2}}}$$

PERIODIC TABLE OF THE ELEMENTS

PERIODIC TABLE OF THE ELEMENTS										KEY										2 He 4.003 Helium									
1 H 1.008 Hydrogen		3 Li 6.941 Lithium		4 Be 9.012 Beryllium		Atomic Number Symbol Standard Atomic Weight Name		79 Au 197.0 Gold		5 B 10.81 Boron		6 C 12.01 Carbon		7 N 14.01 Nitrogen		8 O 16.00 Oxygen		9 F 19.00 Fluorine											
11 Na 22.99 Sodium		12 Mg 24.31 Magnesium								13 Al 26.98 Aluminium		14 Si 28.09 Silicon		15 P 30.97 Phosphorus		16 S 32.07 Sulfur		17 Cl 35.45 Chlorine											
19 K 39.10 Potassium		20 Ca 40.08 Calcium		21 Sc 44.96 Scandium		22 Ti 47.87 Titanium		23 V 50.94 Vanadium		24 Cr 52.00 Chromium		25 Mn 54.94 Manganese		26 Fe 55.85 Iron		27 Co 58.93 Cobalt		28 Ni 58.69 Nickel											
37 Rb 85.47 Rubidium		38 Sr 87.61 Strontium		39 Y 88.91 Yttrium		40 Zr 91.22 Zirconium		41 Nb 92.91 Niobium		42 Mo 95.96 Molybdenum		43 Tc Technetium		44 Ru 101.1 Ruthenium		45 Rh 102.9 Rhodium		46 Pd 106.4 Palladium											
55 Cs 132.9 Caesium		56 Ba 137.3 Barium		57–71 Lanthanoids 89–103 Actinoids		72 Hf 178.5 Hafnium		73 Ta 180.9 Tantalum		74 W 183.9 Tungsten		75 Re 186.2 Rhenium		76 Os 190.2 Osmium		77 Ir 192.2 Iridium		78 Pt 195.1 Platinum											
87 Fr Francium		88 Ra Radium				104 Rf Rutherfordium		105 Db Dubnium		106 Sg Seaborgium		107 Bh Bohrium		108 Hs Hassium		109 Mt Meitnerium		110 Ds Darmstadtium											
										111 Rg Roentgenium		112 Cn Copernicium		113 Nh Nihonium		114 Fl Flerovium		115 Mc Moscovium											
										116 Lv Livermorium		117 Ts Tennessine		118 Og Oganesson															

Lanthanoids

57 La 138.9 Lanthanum	58 Ce 140.1 Cerium	59 Pr 140.9 Praseodymium	60 Nd 144.2 Neodymium	61 Pm Promethium	62 Sm 150.4 Samarium	63 Eu 152.0 Europium	64 Gd 157.3 Gadolinium	65 Tb 158.9 Terbium	66 Dy 162.5 Dysprosium	67 Ho 164.9 Holmium	68 Er 167.3 Erbium	69 Tm 168.9 Thulium	70 Yb 173.1 Ytterbium	71 Lu 175.0 Lutetium
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Actinoids

89 Ac Actinium	90 Th 232.0 Thorium	91 Pa 231.0 Protactinium	92 U 238.0 Uranium	93 Np Neptunium	94 Pu Plutonium	95 Am Americium	96 Cm Curium	97 Bk Berkelium	98 Cf Californium	99 Es Einsteinium	100 Fm Fermium	101 Md Mendelevium	102 No Nobelium	103 Lr Lawrencium
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Standard atomic weights are abridged to four significant figures.

Elements with no reported values in the table have no stable nuclides.

Information on elements with atomic numbers 113 and above is sourced from the International Union of Pure and Applied Chemistry Periodic Table of the Elements (November 2016 version). The International Union of Pure and Applied Chemistry Periodic Table of the Elements (February 2010 version) is the principal source of all other data. Some data may have been modified.



CANDIDATE NUMBER							

2022 FORM V ANNUAL EXAMINATION

Physics

Section I - Multiple Choice

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
 A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word *correct* and drawing an arrow as follows.

A ☒ B ☒ C ☐ D ☐
correct (arrow pointing to B)

**Start
Here** →

1. A ☐ B ☐ C ☐ D ☐
2. A ☐ B ☐ C ☐ D ☐
3. A ☐ B ☐ C ☐ D ☐
4. A ☐ B ☐ C ☐ D ☐
5. A ☐ B ☐ C ☐ D ☐
6. A ☐ B ☐ C ☐ D ☐
7. A ☐ B ☐ C ☐ D ☐
8. A ☐ B ☐ C ☐ D ☐
9. A ☐ B ☐ C ☐ D ☐
10. A ☐ B ☐ C ☐ D ☐
11. A ☐ B ☐ C ☐ D ☐
12. A ☐ B ☐ C ☐ D ☐
13. A ☐ B ☐ C ☐ D ☐
14. A ☐ B ☐ C ☐ D ☐

Back of Multiple-Choice
Answer Sheet



2022 ANNUAL EXAMINATION

PHYSICS

Form V

CRIB

STRUCTURE OF PAPER

Section I: Multiple Choice (14 marks)

Answer all questions on the Multiple-Choice Answer Sheet.

Section II: Extended Response (76 marks)

Answer all parts of the questions in the spaces provided in the Examination Paper.

Board approved calculators may be used

EXAMINATION

Date: Friday 2nd September 8.40am
Duration: 2 hours
Marks: 90

CHECKLIST

In the stapled bundle each boy should have the following:

- Examination sections
- 1 sheet extra writing paper
- Data and Formula Sheets
- Multiple-Choice Answer Sheet

EXAM INSTRUCTIONS

Remove the centre staple and hand in all parts of the examination in a neat bundle.

WRITE YOUR CANDIDATE NUMBER IN THE SPACE PROVIDED AT THE TOP OF EACH SEPARATE SECTION.

There is a Data/Formula sheet included at the end of the paper.

Responses requiring more writing space should be clearly be marked **CONTINUED**. Additional extra writing space is provided at the end of the exam. When the response is continued on extra writing paper it should clearly indicate the question number.

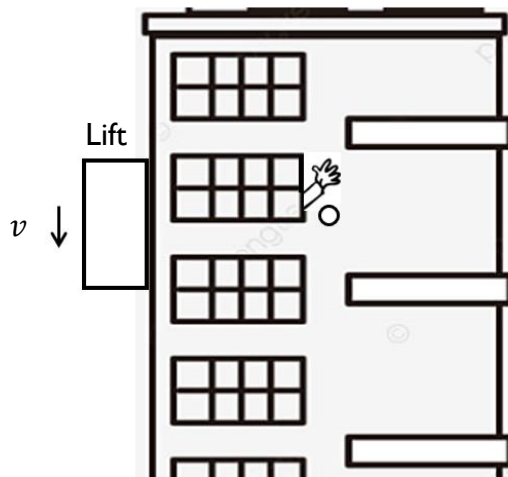
Breakdown of Multiple-Choice responses for the form

		A	B	C	D
Q1	B	2%	84%	13%	1%
Q2	A	57%	5%	13%	25%
Q3	B	3%	93%	3%	0%
Q4	D	6%	9%	17%	68%
Q5	B	3%	75%	3%	18%
Q6	C	0%	9%	89%	2%
Q7	A	69%	22%	6%	1%
Q8	D	7%	16%	8%	69%
Q9	C	2%	10%	43%	44%
Q10	B	8%	81%	3%	8%
Q11	D	14%	9%	35%	42%
Q12	D	0%	0%	0%	100%
Q13	B	6%	76%	8%	10%
Q14	A	50%	26%	16%	6%

SECTION I: MULTIPLE CHOICE

Attempt ALL Questions
Use the Multiple-Choice Answer Sheet.

- 1 A ball is thrown vertically upwards. As it passes its highest point, which of the following quantities will change sign?
- (A) Displacement
(B) Velocity
(C) Acceleration
(D) Force
- 2 A lift is moving down the side of a building at constant speed v . As it passes a window, a ball is dropped from rest.



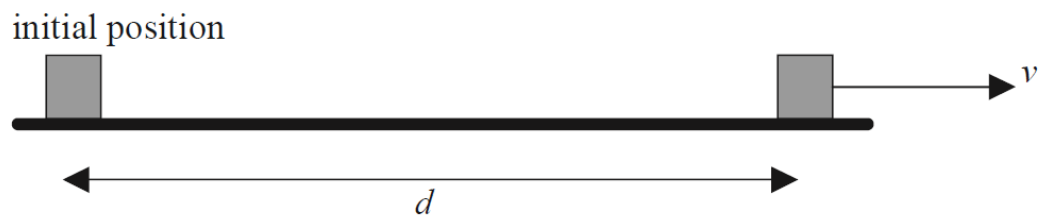
Just after the ball is released, the velocity of the lift relative to the ball will be

- (A) Downwards and decreasing in magnitude.
(B) Downwards and increasing in magnitude.
(C) Upwards and decreasing in magnitude.
(D) Upwards and increasing in magnitude.

- 3 A train accelerates uniformly from rest at a rate of 0.28 m s^{-2} over a distance of 400 m. What time did it take to cover that distance?

(A) 38 s
(B) 53 s
(C) 56 s
(D) 112 s

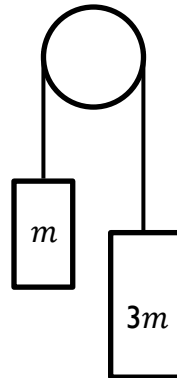
- 4 A box is accelerated from rest at a constant rate. When it is a distance d from its initial position it is moving with a velocity v .



When it was a distance $\frac{d}{2}$ from its initial position its velocity was

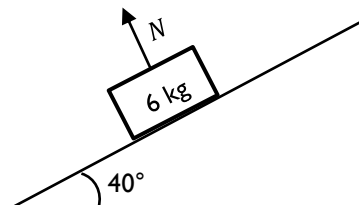
(A) $\frac{v}{4}$
(B) $\frac{v}{2\sqrt{2}}$
(C) $\frac{v}{2}$
(D) $\frac{v}{\sqrt{2}}$

- 5 The diagram shows two masses joined by a lightweight string over a pulley.



What is the magnitude of the acceleration of the system?

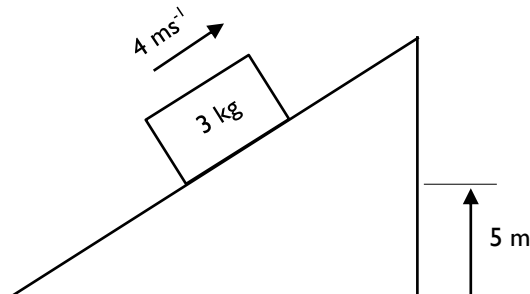
- (A) 0.5 m s^{-2}
(B) 4.9 m s^{-2}
(C) 6.5 m s^{-2}
(D) 9.8 m s^{-2}
- 6 The diagram shows a stationary 6 kg block on a ramp.



What is the magnitude of the normal force on the block?

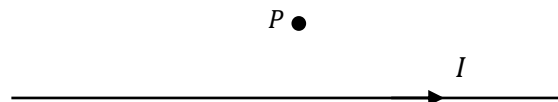
- (A) 4.6 N
(B) 37.8 N
(C) 45.0 N
(D) 58.8 N

- 7 A block is on a frictionless slope, initially 5 m vertically above the ground. It is given an initial velocity of 4 m s^{-1} and slides up the slope before eventually stopping and returning down the slope until it reaches the bottom.



What is the velocity of the block when it reaches the bottom?

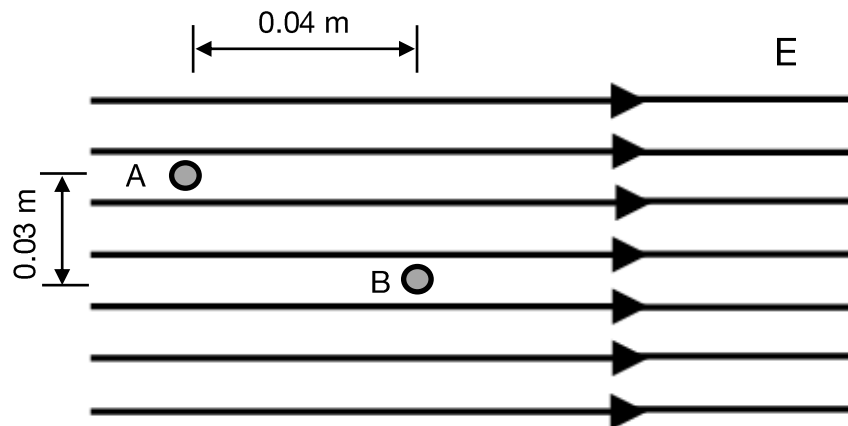
- (A) 10.7 m s^{-1}
(B) 12.1 m s^{-1}
(C) 24 m s^{-1}
(D) 114 m s^{-1}
- 8 A wire carries a current as shown.



If a compass was placed at point P , the compass needle would point

- (A) to the left.
(B) to the right.
(C) into the page.
(D) out of the page.

- 9 A charge of $7.0 \times 10^{-6} \text{ C}$ is moved from position A to position B within a uniform electric field of strength 2000 N C^{-1} .



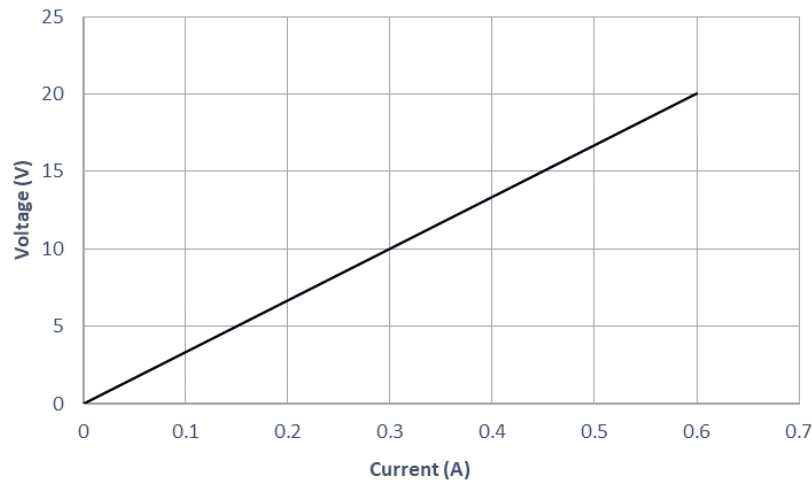
What will be the change in energy of the charge?

- (A) $2.8 \times 10^{-4} \text{ J}$
(B) $4.2 \times 10^{-4} \text{ J}$
(C) $5.6 \times 10^{-4} \text{ J}$
(D) $7.0 \times 10^{-4} \text{ J}$
- 10 A radio wave of frequency 300 MHz would have a wavelength of
- (A) 0.1 m
(B) 1.0 m
(C) 10 m
(D) $1.0 \times 10^6 \text{ m}$
- 11 A light ray that has a wavelength of 600 nm in air travels from the air into glass with refractive index of 1.5.

Determine the frequency of the light ray in the glass.

- (A) $3.3 \times 10^{11} \text{ Hz}$
(B) $5.0 \times 10^{11} \text{ Hz}$
(C) $3.3 \times 10^{14} \text{ Hz}$
(D) $5.0 \times 10^{14} \text{ Hz}$

- 12 The graph below shows the voltage-current characteristics of a resistor.



Calculate the resistance of the resistor.

- (A) $0.03 \, \Omega$
(B) $12 \, \Omega$
(C) $20 \, \Omega$
(D) $33 \, \Omega$
- 13 When X identical resistors are wired in series, their total resistance is 64 times bigger than when they are wired in parallel. How many resistors are there?
- (A) 4
(B) 8
(C) 16
(D) 64
- 14 In a particle accelerator, bunches of high-speed protons travel around a ring, forming a current of 0.58 amps. If there are 3.12×10^7 bunches passing per second, how many protons are in each bunch?
- (A) 1.2×10^{11}
(B) 1.8×10^{11}
(C) 3.6×10^{18}
(D) 3.4×10^{26}

SECTION II: Part A (19 Marks)

Answers

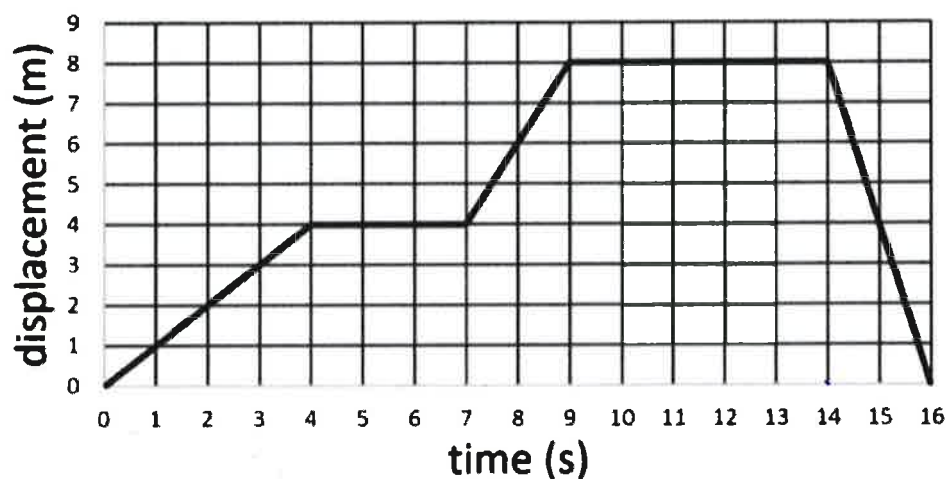
Answer the questions in the spaces provided.
Show all relevant working in questions involving calculations.

CANDIDATE NUMBER							

Question 15 (3 marks)

Marks

The graph below shows the displacement of a remote-controlled car as it moves along a straight line.



- (a) What is the total distance traveled by the car?

16m

.....

.....

1

- (b) What is the maximum speed of the car?

$v = \frac{8}{2} = 4 \text{ ms}^{-1}$

.....

.....

1

- (c) What is the magnitude of the average velocity of the car between 0 and 14 s?

$v_{av} = \frac{8 \text{ m}}{14 \text{ s}} = 0.57 \text{ ms}^{-1}$

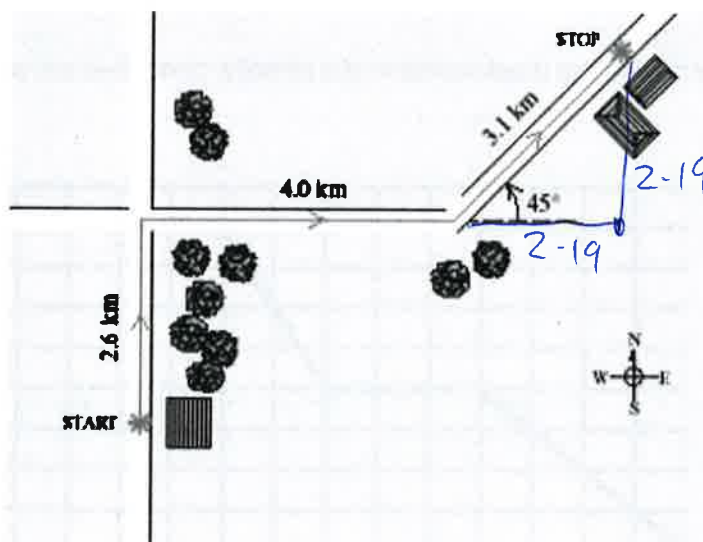
.....

.....

1

Question 16 (4 marks)**Marks**

A car drives from one house to another along the path shown below.



- (a) If the trip takes 20 minutes, calculate the average speed of the car.

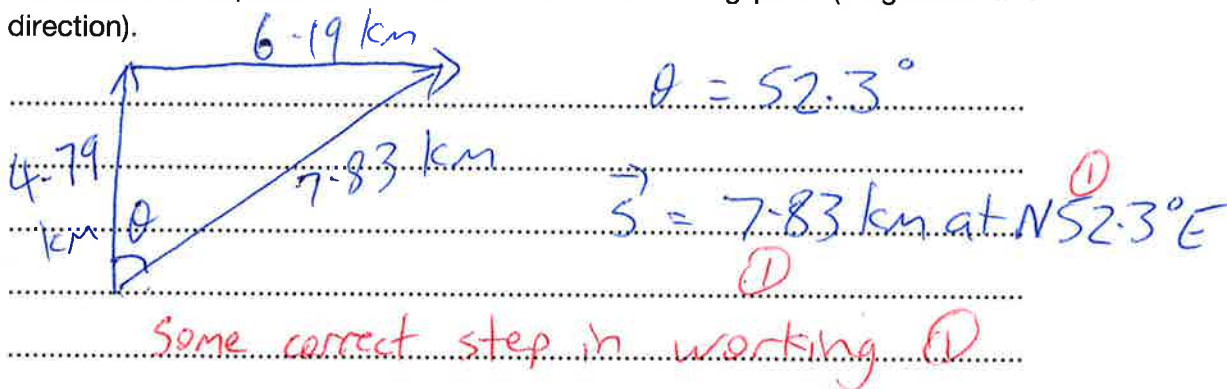
$$\text{av speed} = \frac{d}{t} = \frac{2.6 + 4.0 + 3.1}{20 \times 60} = \frac{9.7 \text{ km}}{1200 \text{ s}}$$

$$= 8.08 \text{ ms}^{-1} \text{ or } 29.1 \text{ kmh}^{-1}$$

either unit is OK

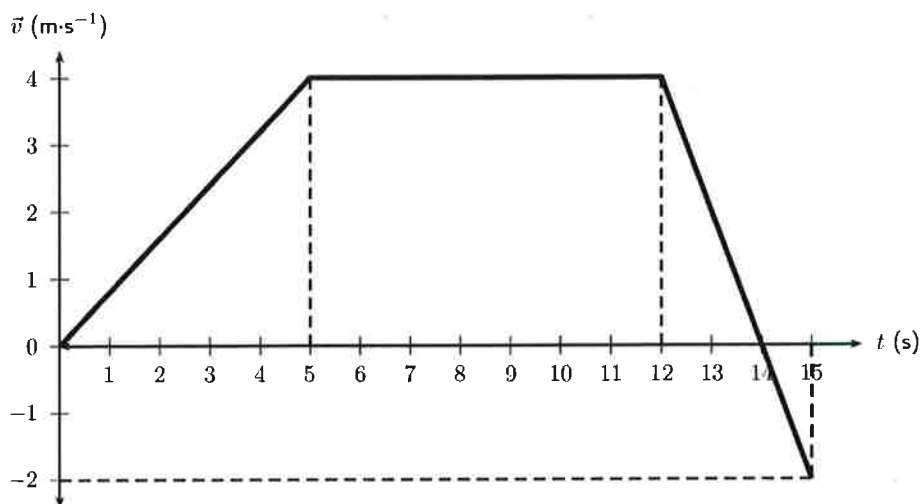
1

- (b) Calculate the displacement of the car from its starting point (magnitude and direction).

**3**

Question 17 (4 marks)**Marks**

A cyclist is moving along a straight road, initially travelling west. The graph of his velocity as a function of time is shown.



- (a) Identify a time at which the cyclist was stationary.

Either 0s or 14s. *Either one ①*

1

- (b) Calculate the magnitude of the cyclist's maximum acceleration.

$$a_{\max} = \frac{4 - (-2)}{3\text{ s}} = 2 \text{ m.s}^{-2}$$

1

- (c) Calculate the displacement at the end of 15 s.

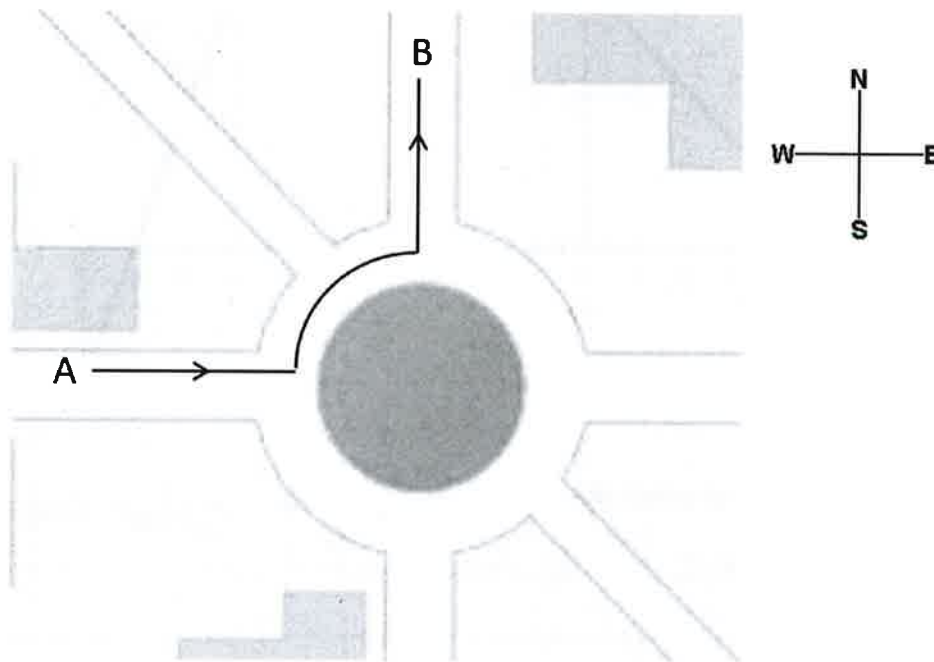
$$\begin{aligned}
 s &= \frac{1}{2} \times 4 \times 5 + 4 \times 7 + \frac{1}{2} \times 4 \times 2 \\
 &\quad - \frac{1}{2} \times 2 \times 1 \quad \text{XNB subtract area below axis} \\
 &= 10 + 28 + 4 - 1 \\
 &= 41 \text{ m west}
 \end{aligned}$$

① ①

2

Question 18 (3 marks)**Marks**

A car uses a roundabout to make a left turn, following the path shown below. The circular section of the car's path has a radius of 16 m and the car maintains a constant speed from A to B.



- (a) If the car takes 4.6 s to complete the circular section of the road, calculate its speed.

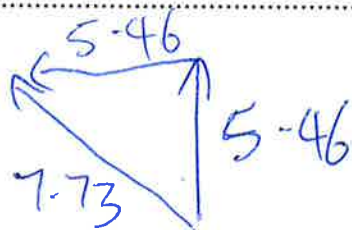
$$d = \frac{1}{4} \times 2\pi \times 16 = 25.1 \text{ m}$$

$$sp = \frac{d}{t} = \frac{25.1}{4.6} = 5.46 \text{ ms}^{-1}$$

1

- (b) Calculate the change in velocity of the car from A to B (magnitude and direction).

$$\Delta \vec{v} = \vec{v}_B - \vec{v}_A = 7.73 \text{ ms}^{-1} \text{ at } N45^\circ W$$

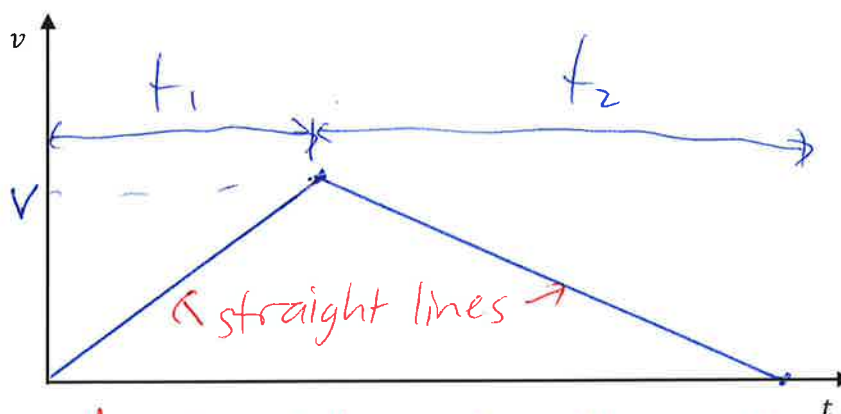
2

Question 19 (5 marks)**Marks**

A plane can accelerate at 3 m s^{-2} from rest down a runway that is 2000 m long. If the plane cannot take off it must slow down again before reaching the end of the runway. It can decelerate at 2 m s^{-2} .

There is a point of no return when the plane cannot stop before the end of the runway if it fails to take off.

- (a) On the axes below, sketch a velocity-time graph for the situation where the plane fails to take off but uses the entire length of the runway (no numerical values required).



t_1 should be noticeably smaller than t_2 .
If they were approximately equal - 1 mark ²

- (b) Calculate the maximum speed reached by the plane in that situation.

$$\begin{aligned}
 v &= 3t_1, & v &= 2t_2 & \left. \begin{array}{l} \text{any correct} \\ \text{relationship between} \end{array} \right\} \\
 \text{so } t_2 &= \frac{3}{2}t_1 & & & \text{link time to displacement times} \\
 s &= \text{area under curve} = \frac{1}{2}vt_1 + \frac{1}{2}vt_2 = 2000 \\
 4000 &= v(t_1 + t_2) = v \cdot \frac{5}{2}t_1 \\
 &= 3t_1 \cdot \frac{5}{2}t_1 = 7.5t_1^2 \\
 t_1 &= 23 \text{ s}, \text{ so } v = 69 \text{ m s}^{-1} \\
 & \text{correct answer}
 \end{aligned}$$

3

SECTION II: Part B (15 Marks)

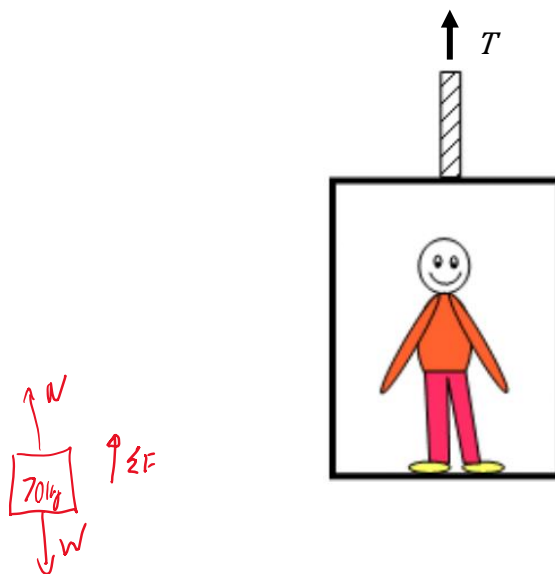
Answer the questions in the spaces provided.
Show all relevant working in questions involving calculations.

CANDIDATE NUMBER							

Question 20 (3 marks)

Marks

A 70 kg man is standing in a lift of mass 800 kg, as shown. The lift is accelerating upwards at 1.5 m s^{-2} .



- (a) Determine the magnitude of the normal force of the lift on the man.

$$\begin{aligned} \uparrow \Sigma F &= N - W = ma \\ N - W + ma &= 0 \\ N &= W - ma = mg - ma = m(g - a) = 70(9.8 - 1.5) = 791 \text{ N} \end{aligned}$$

1

- (b) Determine the magnitude of the tension T in the lift rope.

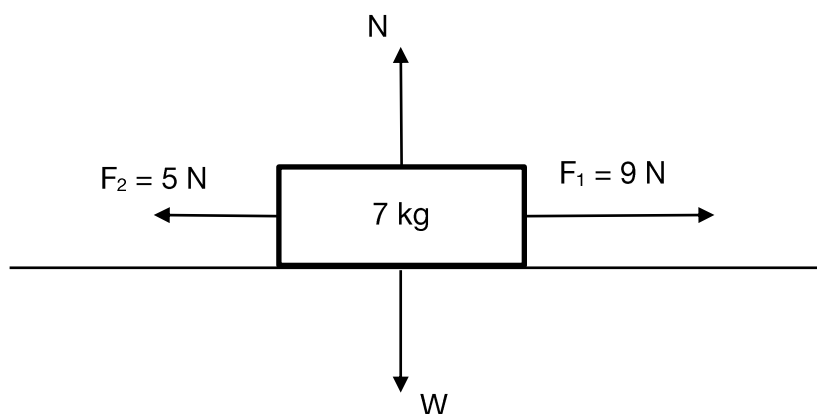
$$\begin{aligned} \uparrow \Sigma F &= T - W = ma \\ T - W + ma &= 0 \\ T &= W - ma = mg - ma = m(g - a) = (800 + 70)(9.8 - 1.5) \\ W + \Sigma F &= 8526 + 1305 = 9831 \text{ N} \end{aligned}$$

2

Question 21 (5 marks)

Marks

A 7 kg object is pulled by an applied force $F_1 = 9 \text{ N}$ and moves to the right. There is a friction force $F_2 = 5 \text{ N}$ acting against its direction of motion.



- (a) Determine the magnitude of the normal force on the block.

Vertically $\Sigma F_y = 0$ so $N = W = 7 \times 9.8 = \boxed{68.6 \text{ N}}$

1

- (b) Determine the acceleration of the block.

$\rightarrow^+ \Sigma F = ma$ $a = \frac{\Sigma F}{m} = \frac{F_1 - F_2}{m} = \frac{9 - 5}{7} = \frac{4}{7}$ ① correct net force
 $= \boxed{0.571 \text{ m/s}^2}$ ① correct answer
 direction ignored 2

- (c) Determine the coefficient of kinetic friction between the block and the surface.

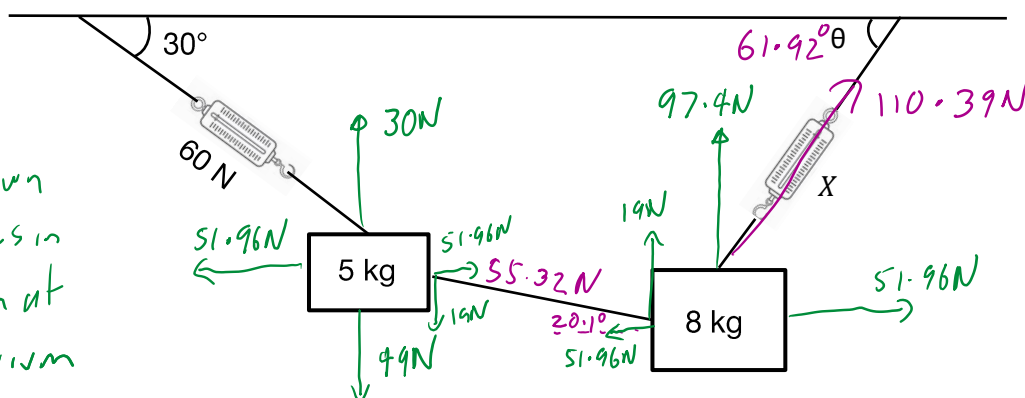
$F_2 = \mu N$ $\mu = \frac{F_2}{N} = \frac{5}{68.6}$ ① correctly identified friction force and correct equation
 $= \boxed{0.073}$ ① correct answer 2

Question 22 (4 marks)

Marks

Two masses are suspended at rest from a ceiling by two spring balances, one of which has unknown tension X . The masses are attached together as pictured below.

Here is
a breakdown
of all forces in
this system at
equilibrium



- (a) Determine the horizontal component of the 60 N force.

$60 \cos 30^\circ$ 51.96 N - correct answer

1

- (b) Determine the angle θ .

For both masses Vertically $\uparrow$ $60 \sin 30^\circ + X \sin \theta = (5+8)g$ - ① correctly balanced
Horizontally $\rightarrow$ $X \cos \theta - 60 \cos 30^\circ = 0$ - ② equations

$X \sin \theta = 13g - 60 \sin 30^\circ = 97.4 \text{ N}$

$X \cos \theta = 60 \cos 30^\circ = 51.96 \text{ N}$

$\frac{X \sin \theta}{X \cos \theta} = \tan \theta = \frac{13g - 60 \sin 30^\circ}{60 \cos 30^\circ} = \frac{127.4 - 30}{51.96} = 1.87$

$\theta = 61.92^\circ$ or 62° - ③ correct answer

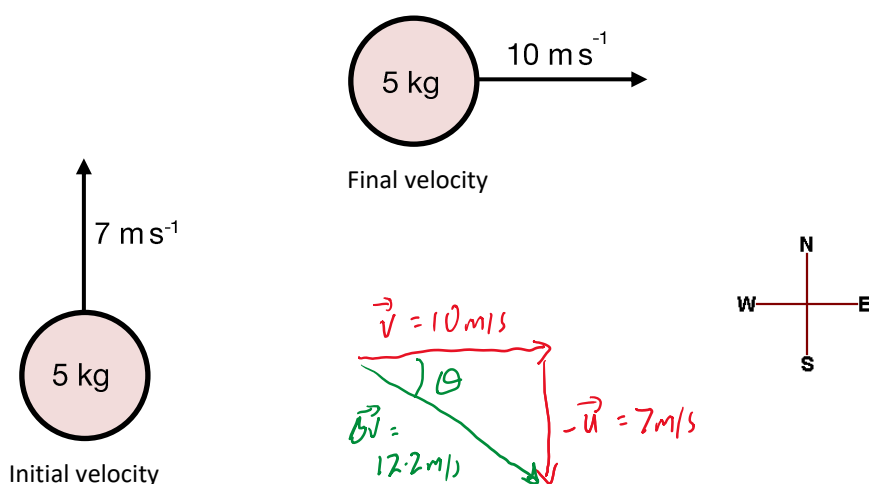
3

Question 23 (3 marks)

Marks

A 5 kg object is travelling initially north at 7 m s^{-1} when a constant force is applied to it for 4 seconds.

After 4 seconds the object is travelling at 10 m s^{-1} east.



Calculate the magnitude and direction of the force that is required to change the object's motion as described.

magnitude $\vec{a} = \frac{\vec{\Delta v}}{t} = \frac{\vec{v} - \vec{u}}{t} = \frac{\sqrt{10^2 + 7^2}}{4} = \frac{12.2}{4} = 3.05 \text{ m/s}^2$ calculate $\vec{\Delta v}$ or $\vec{a}$ using vectors

$\vec{F} = m\vec{a} = 5 \text{ kg} \times 3.05 \text{ m/s}^2 = 15.258 = 15.3 \text{ N}$ correct Force magnitude

direction $\tan \theta = \frac{7}{10}$, $\theta = 35^\circ$ (depicted in diagram) } angle correct 3
 $= 55^\circ \text{ E or } 125^\circ \text{ T or } 6125^\circ$

direction must be unambiguously specific what the angle is specific to

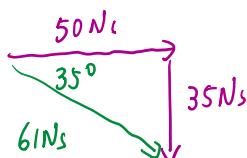
SE is not correct.

• If vector subtraction is not used then no marks can be given

Some solved as a momentum problem

$$\vec{I} = \vec{F}t = \Delta \vec{p} = \vec{p}_f - \vec{p}_i = 61 \text{ Ns}$$

$$F = \frac{61 \text{ Ns}}{4 \text{ s}} = 15.3 \text{ N}$$



SECTION II: Part C (15 Marks)

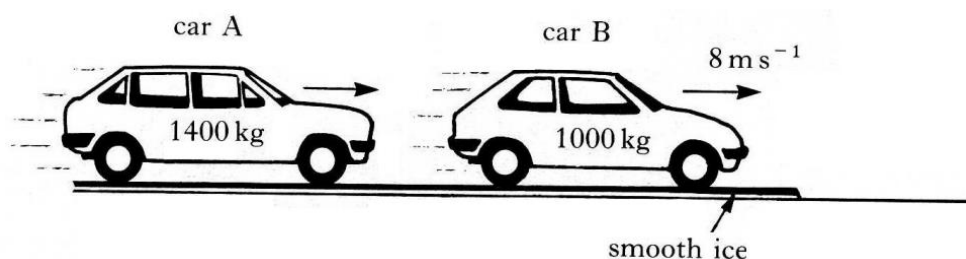
Answer the questions in the spaces provided.
Show all relevant working in questions involving calculations.

CANDIDATE NUMBER							

Question 24 (5 marks)

Marks

Two cars travelling in the same direction but at different speeds, skid on a patch of smooth ice, as shown below.



Car A, of mass 1400 kg, skids straight into the back of Car B, of mass 1000 kg. Immediately before the impact, Car B is moving with a speed of 8 m s^{-1} . The two cars become entangled after the impact and continue to move in the same straight line.

Immediately after the impact, the speed of the entangled cars is 15 m s^{-1} .

- (a) Determine the magnitude of the momentum of Car B just before the collision.

$$p_B(\text{initial}) = m_B u_B = 1000 \text{ kg} \times 8 \text{ m/s} = \underline{8000 \text{ kg} \cdot \text{m/s}}$$

1

- (b) Determine the speed of Car A just before the collision.

$$\begin{aligned} \text{Conservation of Total Momentum: } m_A u_A + m_B u_B &= (m_A + m_B) v \quad (1) \\ \Rightarrow 1400 u_A + 1000 \times 8 &= (1400 + 1000) \times 15 \\ \Rightarrow u_A &= \frac{2400 \times 15 - 8000}{1400} = \underline{20 \text{ m/s}} \quad (1) \end{aligned}$$

2

- (c) After the collision, the cars leave the patch of ice and continue skidding along the rough road. They come to rest in a distance of 20 m after leaving the ice. Calculate the magnitude of the average frictional force acting on the cars as they come to rest.

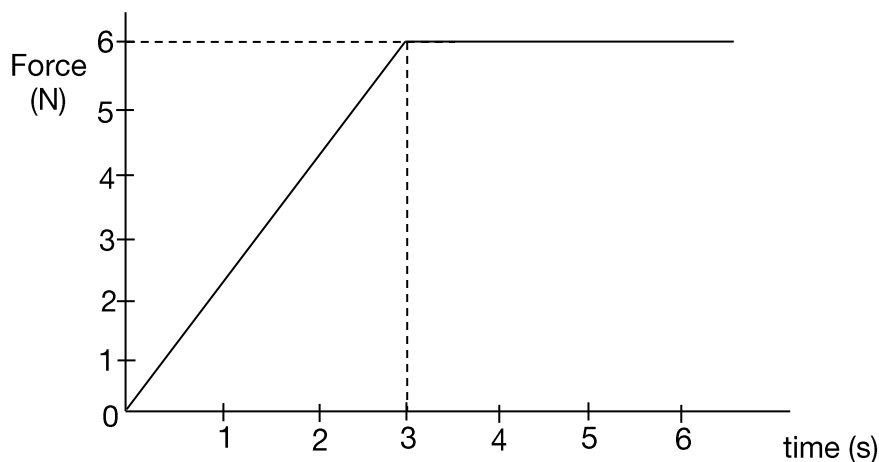
$$\begin{aligned} \Delta K &= \text{work done by friction} = F d \quad (1) \\ \Rightarrow \frac{1}{2} m (v^2 - u^2) &= F d \text{ with } v = 0, u = 15 \text{ m/s and } d = 20 \text{ m} \\ \Rightarrow F &= \frac{\frac{1}{2} \times 2400 \times 15^2}{20} = \underline{13500 \text{ N}} \quad (1) \end{aligned}$$

$$\begin{aligned} \text{OR } F_{\text{net}} &= \text{friction} = ma \text{ with } a = \frac{v^2 - u^2}{2s} = \frac{0^2 - 15^2}{2 \times 20} = -5.625 \text{ m/s}^2 \quad (1) \\ \text{So friction} &= (1400 + 1000) \times 5.625 = \underline{13500 \text{ N}} \quad (1) \end{aligned}$$

Question 25 (4 marks)

Marks

The graph below shows how the force on a 2.4 kg object varies with time. The object's velocity was initially 10 m s^{-1} east and the force was applied in a westerly direction.



- (a) Calculate the object's change in momentum during the first 3 seconds (magnitude and direction).

$\Delta p = \text{impulse}$. Here, $\Delta p = -Ft$ (as the force is acting in the opposite direction to the motion)

So $\Delta p = -\text{area under graph} = -\frac{1}{2} \times 3 \times 6 = -9 \text{ kg.m/s East}$ (or 9 kg.m/s West)

2

- (b) Determine the time at which the object is at rest.

$\Delta p = -9 - [6 \times (t - 3)]$ $\Leftrightarrow m(v - u) = -9 - 6t + 18$ with $v = 0$
 $\Leftrightarrow -2.4 \times 10 = 9 - 6t$ $m = 2.4 \text{ kg}$
 $\Leftrightarrow t = \frac{-33}{-6} = 5.5 \text{ s}$

for the first 3 seconds time after 3 seconds

2

OR At $t = 3 \text{ s}$, the momentum has dropped by 9 kg.m/s so only $24 - 9 = 15 \text{ kg.m/s}$ remains

initial momentum: $2.4 \times 10 = 24 \text{ kg.m/s}$

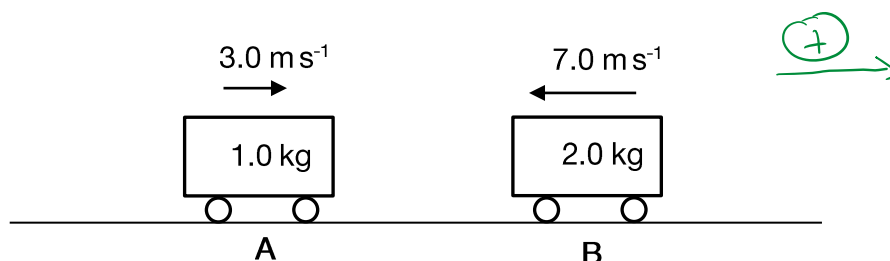
area under graph: $15 = 6 \times t \Rightarrow t = \frac{15}{6} = 2.5 \text{ s after } t = 3 \text{ s}$

$\Rightarrow T = 3 + 2.5 = 5.5 \text{ s}$

Question 26 (3 marks)

Marks

Trolley A of mass 1.0 kg is moving with a velocity of 3.0 m s^{-1} towards the right. It collides with trolley B of mass 2.0 kg moving left with a velocity of 7.0 m s^{-1} .



3 methods

If the collision is perfectly elastic, what is the final velocity of the trolley A?

• Centre of mass: $v_A = 2v_{CM} - u_A = 2 \times \left[\frac{1 \times 3 + 2 \times (-7)}{1+2} \right] - 3$

$\Rightarrow v_A = 2 \times (-3.67) - 3 = -10.33 \text{ m/s right (i.e. to the left)}$

OR

• Conservation of Total Momentum: $3 \times 1 + 2 \times (-7) = 1 \times v_A + 2 \times v_B \Rightarrow v_A + 2v_B = -11$ (Eq. A) ①
 then relative velocity: $u_A - u_B = v_B - v_A \Rightarrow 3 - (-7) = v_B - v_A \Rightarrow v_B - v_A = 10$ (Eq. B) ①
 (B) into (A) $\Rightarrow v_A + 2 \times (10 + v_A) = -11 \Rightarrow 3v_A = -31$
 $\Rightarrow v_A = -\frac{31}{3} = -10.33 \text{ m/s right}$ ①

OR • CTM $\Rightarrow v_A + 2v_B = -11$ (Eq. A) ①

Conservation of Total kinetic energy: $\frac{1}{2} \times 1 \times 3^2 + \frac{1}{2} \times 2 \times (-7)^2 = \frac{1}{2} \times 1 \times v_A^2 + \frac{1}{2} \times 2 \times v_B^2$

$\Rightarrow 107 = v_A^2 + 2v_B^2$ (Eq. C)

(A) into (C) $\Rightarrow 107 = v_A^2 + 2 \times \left(\frac{-11 - v_A}{2} \right)^2 \Rightarrow 3v_A^2 + 22v_A - 93 = 0$ } ①

either $v_A = 3 \text{ m/s}$ and $v_B = -7 \text{ m/s}$
 physically impossible

or $v_A = -10.33 \text{ m/s}$ and $v_B = -0.33 \text{ m/s}$
 ①



very common mistake:

using the values of the speeds for the values of the masses!

e.g. writing $-11 = 3v_A - 7v_B$

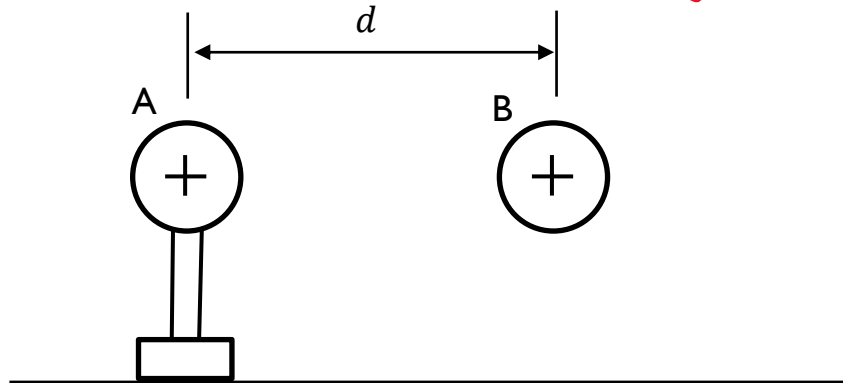
↑ ↑
 these are the initial velocities
 of A and B, not their masses

Question 27 (3 marks)

Marks

A positively charged particle A is fixed on a stand. Another positive charge B is held a distance d away it, as shown. Assume no external forces act on the particles.

→ ⚠ Gravity does NOT act here !



At $t = 0$, particle B is released. Describe what changes will subsequently occur to the motion of particle B and the force between the particles, and justify your answer.

{ 1 mark: Description of initial and subsequent motion (thorough)
 1 mark: Description of initial and subsequent force (thorough)
 1 mark: Reasons provided (just stating $F_e = k \frac{q_A q_B}{d^2}$ is not enough)

- like charges repel so force on B is away from A, with a decreasing magnitude as d increases because $F_e = k \frac{q_A q_B}{d^2}$
 (F is inversely proportional to the square of the distance)
- B accelerates away from A but at a decreasing rate,
 as $F = ma$ (F and a are proportional)

3

⚠ lots of contradictions, e.g. "the decreasing force acting on B means it will accelerate at a lower and lower rate" } correct
 and therefore, B will slow down " } incorrect!

Misconception:

⚠ The fact that B moves away from A has nothing to do with the fact that A is fixed. It would also happen if A was free to move. And it is not doubling the force on B.

B will continue to see its speed increase but at a lower and lower rate

Crib

SECTION II: Part D (15 Marks)

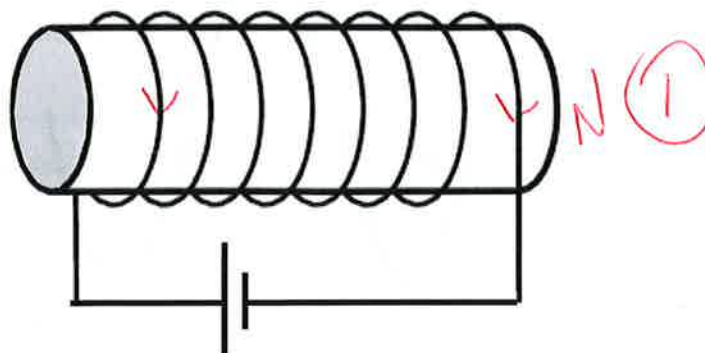
Answer the questions in the spaces provided.
Show all relevant working in questions involving calculations.

CANDIDATE NUMBER								

Question 28 (3 marks)

Marks

A coil of wire is wrapped around a hollow cardboard tube, as shown. A power supply is connected to the coil and the current creates an electromagnet.



The coil carries a current of 6 A and has a length of 15 cm, with 40 turns of wire.

- (a) Determine the strength of the magnetic field inside the coil.

$$B = \mu_0 n I$$
$$= 4\pi \times 10^{-7} \times 40 \times 6 = 0.002 \text{ T}$$

① 0.15

①

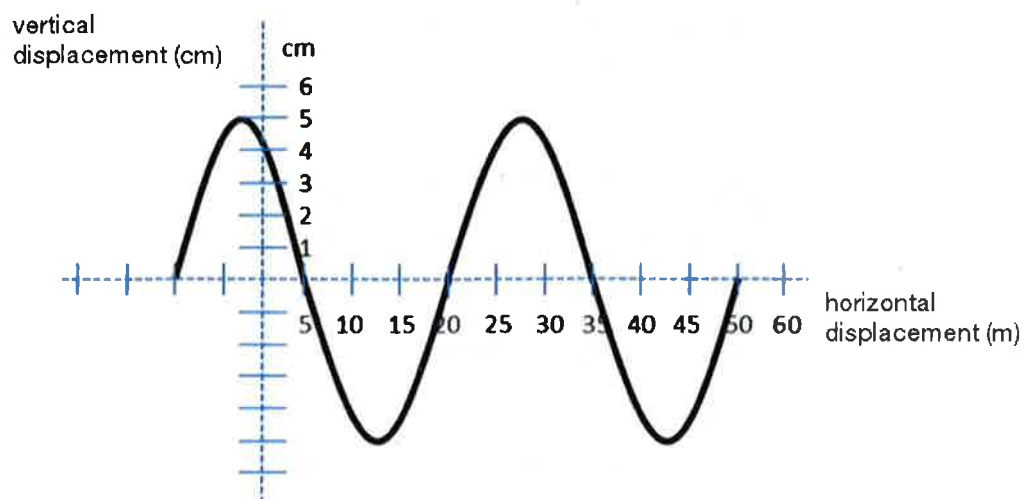
2

- (b) Indicate on the diagram with the letter N the position of the North pole of the electromagnet.

1

Question 29 (4 marks)**Marks**

The graph below shows vertical displacement versus horizontal displacement for a wave travelling to the right at a speed of 10 m s^{-1} .



From the information in the graph, determine:

- (a) the wavelength of the wave.

30 m

1

- (b) the amplitude of the wave.

0.05

1

- (c) the period of the wave.

$$v = f\lambda = \frac{\lambda}{T}$$

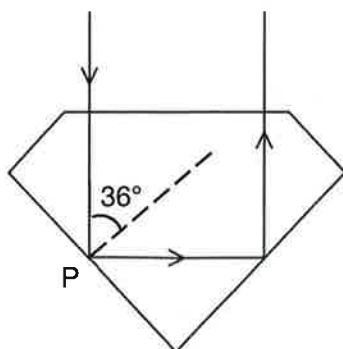
$$T = \frac{\lambda}{v} = \frac{30}{10}$$

$$= 3 \text{ sec.}$$

2

Question 30 (4 marks)**Marks**

A ray of light enters a gemstone as shown below. It reflects twice within the gemstone, then exits.



- (a) Show that the refractive index of the gemstone must be at least 1.7. Justify your answer.

Assume $\theta_c = 36^\circ$

$$\frac{\sin 36^\circ}{\sin 90^\circ} = \frac{n_{\text{air}}}{n_{\text{gemstone}}} = \frac{1}{n_{\text{gemstone}}}$$

$$\therefore n_{\text{gemstone}} = \frac{1}{\sin 36^\circ} = 1.7$$

(2)

We know $36^\circ \geq \theta_{\text{critical}}$ $\therefore n_{\text{gemstone}} \geq 1.7$

(1)

- (b) Assuming that the refractive index of the gemstone is 1.8, determine the speed of the light within the gemstone.

$$v = \frac{c}{n} = \frac{3 \times 10^8}{1.8} = 1.67 \times 10^8 \text{ m/s}$$

(1)

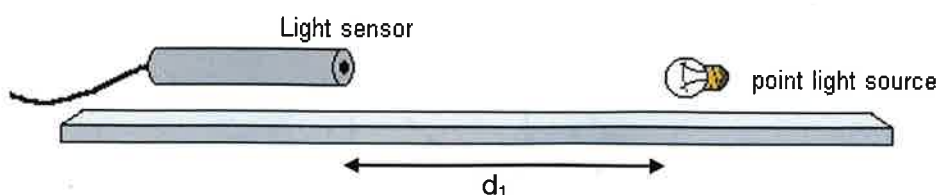
or

If $n_{\text{gemstone}} < 1.7$ then total internal reflection wouldn't occur.

(1)

Question 31 (2 marks)**Marks**

A light sensor which measures the intensity of light is placed a distance $d_1 = 15 \text{ cm}$ from a point light source, as shown.

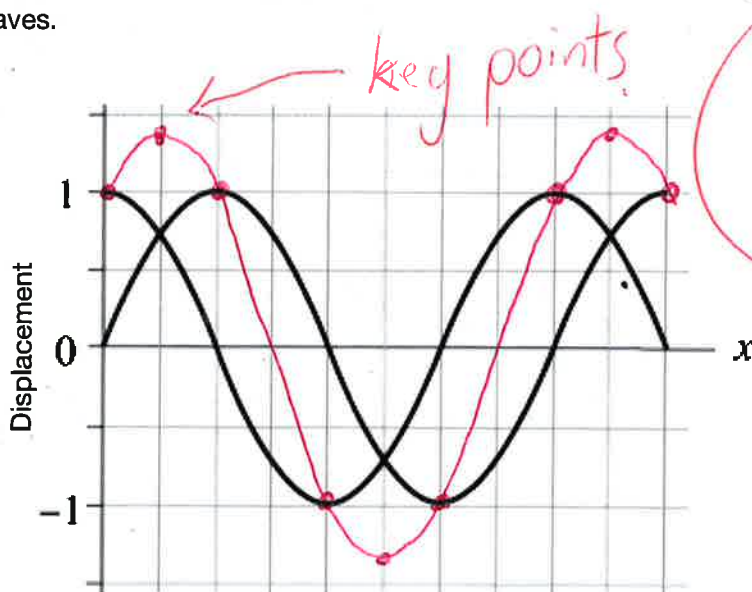


If the intensity at that distance is 2.0 W m^{-2} , determine the distance d_2 at which the intensity will be 1.5 W m^{-2} .

$$\begin{aligned} I_1 r_1^2 &= I_2 r_2^2 \\ 2 \times 0.15^2 &= 1.5 \times r_2^2 \quad (1) \\ r_2^2 &= 0.03 \\ r_2 &= 0.173 \text{ m} = 17.3 \text{ cm} \quad (1) \end{aligned}$$

2**Question 32** (2 marks)

Two waves are depicted on the graph below. Draw on the graph the superposition of the two waves.



Any mistakes
-1 each.

2

AAH

				C	R	I	B
CANDIDATE NUMBER							

Marks

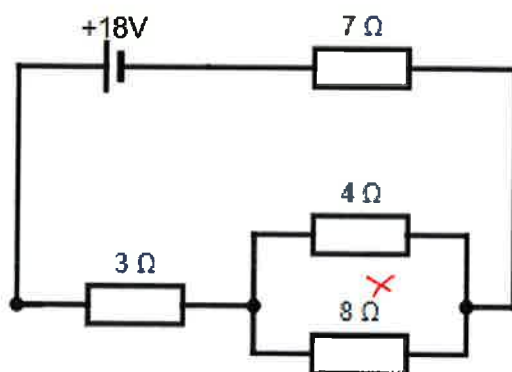
$$R_{TOT} = 8\Omega \quad \therefore I = 16/8 = \underline{2A} \quad \text{--- (A)} \\ \therefore V_1 = 5 \times 2 = 10V \quad \text{--- (1)}$$

$$A_1 = 24$$

$V_1 = 10V$

Question 34 (5 marks)**Marks**

Consider the circuit shown below.



$$\frac{1}{R_x} = \frac{1}{4} + \frac{1}{8}$$

$$\therefore R_x = \frac{8}{3} = 2.67\Omega$$

Calculate:

- (a) The total resistance of the circuit.

$$= 3 + 7 + R_x \quad R_x = 2.67\Omega$$

$$= 12.67\Omega$$

2

- (b) The current through the 3 Ω resistor.

$$= 18 / R_{\text{total}} = 18 / 12.67 = 1.421\text{ A}$$

1

- (c) The voltage across the 8 Ω resistor.

$$= 8 \times I_b = 8 \times 1.421$$

$$= 18 - (10 \times I_b) = 18 - (10 \times 1.421)$$

1

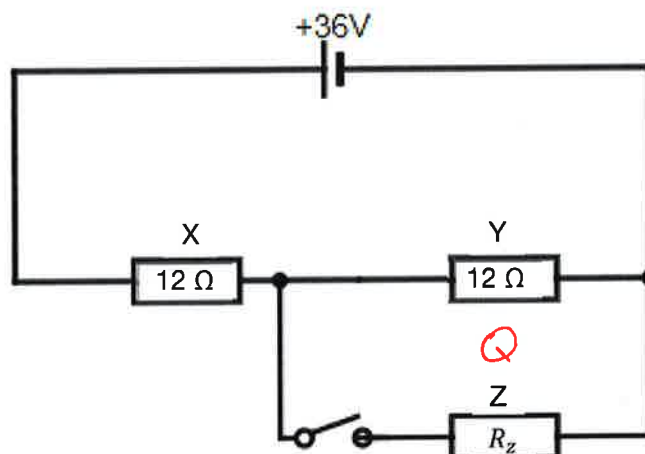
- (d) The current through the 4 Ω resistor.

$$= \frac{3.79}{4} = 0.948\text{ A}$$

1

Question 35 (5 marks)**Marks**

Consider the circuit shown below.



- (a) Calculate the amount of power used by Resistor X when the switch is open.

NB $P_{TOT} = 54W$
 (1) max

$I = 36/24 = 1.5A$ (or $V_X = 18V \dots$)
 (1) or (1)
 $\therefore P_X = I^2 R = 1.5^2 \times 12 = 27W$ — (2)

2

- (b) When the switch is closed, the total power consumption of the whole circuit increases by a factor of $\frac{5}{3}$. Calculate the resistance R_Z of Resistor Z.

(1) $\rightarrow$ IF NO MARKS ELSEWHERE!
 $P = \frac{5}{3} \times 27 \times 2 = 90W$ ($\rightarrow$ or $I = 2.5A$)
 (1) or (1)
 $\therefore R = V^2/P = 14.4\Omega$ $\therefore R_Q = 2.4\Omega$ — (2)
 $\frac{1}{2.4} = \frac{1}{12} + \frac{1}{R_Z}$ $\therefore R_Z = 3\Omega$ — (3)

3

END OF EXAMINATION